

1. Purpose

The Leslie Population Dynamics App is a browser-based simulation tool for stochastic age-structured population modeling. It was designed for species whose population dynamics can be represented clearly with yearly age classes, age-specific fecundity, age-specific survival, and density dependence acting through emigration above carrying capacity.

This version of the program supports:

- up to 20 yearly age classes
- a user-defined first reproductive age class
- reproduction through all later ages
- extinction defined by total population size reaching zero
- single-location mode
- metapopulation mode with multiple habitat patches
- single-species simulations
- two-species isolated simulations
- two-species coexisting simulations with a joint effective carrying capacity
- environmental stochasticity
- random catastrophe years with optional reproductive failure
- demographic stochasticity
- up to 1000 replicate simulations
- summary plots and CSV export

By default, the Leslie-style schedules are initialized with zeros so that a user can enter a custom system directly without first clearing example values.

2. Main Biological Assumptions

The current model assumes:

- time advances in yearly steps
- no individual survives beyond age class 20
- reproduction starts at the first reproductive age class selected by the user
- all older age classes remain reproductive
- annual fecundity and survival vary stochastically around user-defined means

- carrying capacity may vary stochastically from year to year
- catastrophe years may occur randomly around a user-defined average interval
- in metapopulation mode, each patch has its own carrying capacity and can exchange migrants with other patches
- density dependence does not directly alter fecundity or mortality
- carrying capacity is interpreted from reproductive individuals only, not juveniles
- when projected reproductive abundance exceeds carrying capacity, surplus reproductive individuals leave through emigration

These assumptions make the app especially useful when a transparent Leslie-type age-structured model is biologically appropriate.

3. Launching the Program

The app is stored in the folder:

`E:\LESLIE\leslie\_app`

To launch it:

1. Open the folder.
2. Double-click `launch\_app.bat`.

You can also launch it from PowerShell:

```
cd E:\LESLIE\leslie_app
.\launch_app.bat
```

If needed, you may also run:

```
cd E:\LESLIE\leslie_app
.\.venv\Scripts\python -m streamlit run app.py
```

When the app starts, Streamlit will open the web interface in your browser.

4. Simulation Modes

The app currently supports three modes.

4.1 Single

One species is simulated with one carrying-capacity process.

4.2 Two Isolated

Two species are simulated separately. Each species receives its own annual carrying-capacity draw from the same `K` distribution. This mode is useful for comparing two species or two parameter sets without interaction.

4.3 Two Coexisting

Two species are simulated together under a shared effective carrying capacity. In this mode, both species contribute to crowding, and surplus individuals are removed through emigration.

4A. Spatial Modes

The program now supports two spatial scales.

4A.1 Single Location

This is the original one-patch mode. The entire population is modeled as one local population.

4A.2 Metapopulation

In metapopulation mode, the user specifies two or more habitat patches. Each patch runs its own local age-structured dynamics, and emigrants can move among patches through a movement matrix. This allows extinction in one patch to be offset by immigration from another patch.

5. Inputs

5.1 General Settings

- `Simulation mode`
- `Spatial mode`
- `Years`
- `Replicates`
- `Random seed`

`Years` can range from 2 to 1000.

`Replicates` can range from 1 to 1000.

In metapopulation mode, the number of habitat patches must also be specified.

5.1A Patch Settings

When metapopulation mode is selected, the app provides a patch table with:

- `Patch name`
- `Mean K`
- `K SD`
- `Quality`
- `Catastrophe interval`

Each patch has its own carrying capacity process.

If patch-specific catastrophe timing is enabled, each patch also uses its own catastrophe interval for independent catastrophe draws.

5.2 Carrying Capacity

- `Mean K`
- `K SD`
- `Use heavy-tailed annual shocks`
- `Heavy-tail degrees of freedom`
- `Species 1 load weight`
- `Species 2 load weight`

In coexisting mode, the species weights determine how strongly each species contributes to the shared carrying-capacity burden.

In the current version, `K` is evaluated against the reproductive part of the population. Juveniles are still simulated and can disperse, but they do not count directly toward carrying-capacity exceedance.

5.3 Catastrophes

- `Enable catastrophic reproductive failure years`
- `Average catastrophe interval (years)`
- `Use patch-specific catastrophe timing`

If catastrophes are enabled, the program draws catastrophe years stochastically. The yearly probability is set so that the long-term mean spacing between catastrophe years is close to the value chosen by the user.

If `Use patch-specific catastrophe timing` is not enabled, the same catastrophe year affects the whole system.

If `Use patch-specific catastrophe timing` is enabled in metapopulation mode, each patch draws catastrophe years independently around its own average interval. This creates the asynchronous local failures that are often essential for rescue effects and recolonization dynamics.

Each species also has a separate switch:

- `Reproduction fails in catastrophe years`

This means a catastrophe can affect one species only or both species simultaneously.

5.4 Species-Specific Inputs

For each species, the user can specify:

- `Species name`
- `First reproductive age class`
- `Fecundity SD`
- `Survival SD`
- `Reproduction fails in catastrophe years`
- `Initial population` in each age class
- `Mean fecundity` in each age class
- `Mean survival to next age`
- `Migration weight` in each age class
- `Baseline emigration` in each age class
- `Density emigration` in each age class

In metapopulation mode, each species also has:

- an initial population table by patch
- a patch-to-patch movement matrix

To make editing easier, the species section is split into two tables:

- `Local demographic schedule`
- `Age-specific emigration schedule`

This avoids pushing the density-emigration values to the far-right edge of one wide table.

5.5 Meaning of the Editable Table

Each row represents one yearly age class.

- `Age class`: fixed identifier from 1 to 20
- `Initial population`: starting number of individuals in that age class
- `Mean fecundity`: expected birth production from individuals of that age class
- `Mean survival to next age`: survival from the current age class to the next
- `Migration weight`: relative tendency of individuals in that age class to emigrate if the population exceeds carrying capacity

The last age class has no survival-to-next-age transition because age class 20 is terminal.

5.6 Meaning of Migration Parameters

The program uses three migration-related inputs, and they do not mean the same thing.

- `Migration weight`: a unitless relative weight used when the local projected population is above carrying capacity and the model must decide which age classes contribute the most overflow emigrants
- `Baseline emigration`: a per-capita annual probability of leaving a patch even when density is not unusually high
- `Density emigration`: an additional per-capita annual probability of leaving a patch when crowding is high

This means:

- `Migration weight` is not itself a probability
- `Baseline emigration` and `Density emigration` are probabilities on a 0 to 1 scale
- these settings describe emigration, not immigration

Immigration is generated indirectly when emigrants are assigned to destination patches through the movement matrix.

5.7 Movement Matrix in Metapopulation Mode

In metapopulation mode, emigrants are redistributed among patches by a species-specific movement matrix.

- rows represent source patches
- columns represent destination patches
- each cell gives the expected proportion of emigrants from the source patch that move to the destination patch

For example, if the row for `Patch 1` is:

- `Patch 1 -> Patch 1 = 0`
- `Patch 1 -> Patch 2 = 0.5`
- `Patch 1 -> Patch 3 = 0.5`

then emigrants leaving `Patch 1` are split randomly, with an expected half going to `Patch 2` and half to `Patch 3`.

The realized number of immigrants in a given year is stochastic, so exact values vary among replicates.

6. How the Model Works

For each year of a simulation:

1. The program draws annual stochastic fecundity values around the age-specific fecundity means.
2. The program draws annual stochastic survival values around the age-specific survival means.
3. If a catastrophe occurs and a species is marked as affected, births are set to zero for that species in that year.
4. New births are generated.
5. Survivors advance to the next age class.
6. The next-year population is built before density regulation.
7. Annual carrying capacity is drawn.
8. If projected abundance exceeds carrying capacity, surplus individuals are removed through emigration.

In metapopulation mode, the annual order is extended:

1. local survival
2. local reproduction
3. catastrophe effect
4. local projection within each patch
5. emigration based on age-specific baseline and density-dependent rates
6. redistribution among destination patches using the movement matrix
7. occupancy update at patch and metapopulation levels

6.1 Stochastic Components

The present implementation uses:

- lognormal variation for fecundity
- beta variation for survival
- lognormal variation for carrying capacity
- Poisson births
- binomial survival transitions

In practical terms, the stochastic models are:

- `Mean fecundity by age`: user-defined deterministic mean values
- `Annual fecundity around the mean`: lognormal environmental variation
- `Realized births`: Poisson draws from the annual fecundity outcome
- `Mean survival to next age`: user-defined deterministic mean values
- `Annual survival around the mean`: beta variation constrained to the 0 to 1 interval
- `Realized survivors`: binomial transitions from one age class to the next
- `Mean K`: user-defined deterministic mean carrying capacity
- `Annual K around the mean`: lognormal environmental variation
- `Catastrophe timing`: Bernoulli yearly event process with probability chosen from the average catastrophe interval
- `Patch-specific catastrophe timing`: independent Bernoulli yearly event process in each patch when that option is enabled

This means the program does not use one single distribution for all parameters. Different distributions are used because the parameters have different biological meanings and mathematical constraints.

- fecundity and carrying capacity must remain positive, so lognormal variation is used
- survival must remain between 0 and 1, so beta variation is used
- births are counts, so Poisson draws are used
- survival outcomes are counts of survivors from a known number of individuals, so binomial draws are used

Why fecundity and carrying capacity use lognormal variation

Fecundity and carrying capacity are positive-valued biological quantities. In the model they can be small, moderate, or large, but they should not become negative after an annual environmental draw. A normal distribution would sometimes produce negative fecundity or negative carrying capacity when the standard deviation is large relative to the mean, which would have no biological interpretation.

A lognormal distribution avoids this problem because its draws are always non-negative. It is also biologically useful because environmental effects often act multiplicatively rather than additively. For

example, a favorable year might increase reproduction or effective habitat capacity by a percentage, while a poor year might reduce it by a percentage. This produces asymmetric variation: very high years are possible, but values cannot drop below zero. In LESLIE, user-entered means and standard deviations are converted internally into lognormal parameters so the simulated annual values remain centered around the requested scale.

Why survival uses beta variation

Survival is a probability. It must stay between 0 and 1. The beta distribution is designed for values on this interval, so it is a natural choice for annual variation in survival probabilities. It lets survival fluctuate around the user-defined mean while respecting the biological limits that survival cannot be negative and cannot exceed certainty.

This is different from the realized number of survivors. First, the program draws an annual survival probability for each age-specific transition. Then it applies that probability to the number of individuals currently in the age class.

Why births use a Poisson draw

Births are count data. After the annual fecundity value is drawn, each age class has an expected number of offspring equal to the number of individuals in that age class multiplied by that year's fecundity. The realized number of offspring is then drawn from a Poisson distribution around that expectation.

The Poisson distribution is appropriate here because it represents random variation in counts around an expected rate. This step adds demographic stochasticity: two replicate populations with the same annual fecundity and the same number of reproductive individuals may still produce different realized numbers of births.

What "survival transition" means, and why it varies binomially

A survival transition is the movement of individuals from one age class to the next over one yearly time step. For example, the transition $3 \rightarrow 4$ means individuals that are in age class 3 at the start of the year and are alive in age class 4 at the start of the next year. The final age class has no transition to a later class because age class 20 is terminal in the current model.

The transition varies binomially because each individual in the starting age class has two possible outcomes during that year: survive into the next age class or fail to survive. If there are N individuals in an age class and the annual survival probability for that transition is p , the number that survive is drawn from a binomial distribution with N trials and success probability p . This keeps realized survivors as whole individuals and prevents the model from carrying fractional animals between age classes.

6.1A Heavy-Tailed Annual Shocks

`Use heavy-tailed annual shocks` is an option that makes annual environmental deviations more extreme than under a standard bell-shaped process.

In plain terms:

- without this option, most annual fecundity and carrying-capacity values stay relatively close to their means
- with this option, rare very poor years or very favorable years become more likely

This option does not itself create catastrophes. Catastrophes are handled separately by the catastrophe module, which can force reproductive failure in selected years.

6.1B Shared Versus Patch-Specific Catastrophes

The catastrophe module can now work in two ways.

- `Shared catastrophe timing`: one random catastrophe draw is made for the whole system in a given year
- `Patch-specific catastrophe timing`: each patch makes its own random catastrophe draw in a given year

Patch-specific timing is usually more appropriate for metapopulation questions when the user wants local extinctions and reproduction failures to be asynchronous rather than system-wide.

The catastrophe module is separate from these stochastic processes. It creates discrete years of reproductive failure rather than simply broader annual noise.

6.2 Density Dependence Through Emigration

This program does not implement density dependence by directly lowering fecundity or survival. Instead, if the projected population is larger than the annual carrying capacity, the surplus is removed from the population as emigrants.

In the current implementation, the carrying-capacity comparison is based on reproductive individuals only. Juveniles do not directly count toward `K`, although they remain part of the simulated age structure and can still disperse according to their migration settings.

This feature makes the program suitable for source-sink style interpretations in which excess individuals leave the local population instead of being immediately converted into extra mortality or reduced breeding success.

6.3 Reproductive Population Versus Total Population

The program tracks both:

- `Total population`
- `Reproductive population`

Extinction is determined only by ``Total population = 0``.

The reproductive population is reported separately because it can be a more stable and biologically informative census for many small vertebrates, especially when annual newborn numbers fluctuate strongly.

7. Outputs

After a run, the app provides:

- mean total population trajectory
- SD of total population trajectory
- mean reproductive population trajectory
- SD of reproductive population trajectory
- persistence probability through time
- extinction probability through time
- catastrophe probability through time
- reproduction-failure probability through time
- deterministic Leslie lambda
- realized arithmetic and geometric lambda values
- occupancy proportion through time
- occupied patch counts through time
- patch summary table
- extinction-time histogram
- extinction summary table
- trajectory summary table
- CSV downloads

7.1 Extinction Summary

The summary table includes:

- extinction probability
- survival probability
- mean extinction year
- SD of extinction year
- median extinction year
- restricted mean persistence

- mean number of catastrophe years
- mean number of reproduction-failure years

`Restricted mean persistence` is the average of the extinction year truncated at the chosen time horizon. It is useful when some replicates survive through the final year.

7.2 Lambda and Growth Summary

The app also reports growth-rate metrics.

- `Deterministic Leslie lambda`: the dominant eigenvalue of the mean Leslie matrix built from the entered fecundity and survival coefficients
- `Realized arithmetic lambda`: the average annual ratio $N(t+1) / N(t)$ across stochastic simulated intervals with nonzero denominator
- `Realized geometric lambda`: the geometric mean annual growth rate across stochastic simulated intervals

These are reported both for:

- total population
- reproductive population

The deterministic Leslie lambda represents the matrix-based expectation from the entered coefficients. The realized lambdas reflect the actual simulated growth under stochasticity, catastrophes, emigration, and carrying-capacity effects.

8. Advantages of This Program

This program has several important strengths.

8.1 Transparency

The model is explicit and easy to interpret. The main components are age-specific fecundity, age-specific survival, carrying capacity, and emigration. This makes it easier to understand how parameter changes influence the output.

8.2 Custom Fit to Leslie-Type Questions

The program is directly built around a Leslie-style age-structured framework. It is therefore especially convenient when the biological question is focused on:

- age-specific fecundity

- age-specific survival
- annual environmental variation
- carrying capacity variation
- replicate-based extinction and persistence summaries

8.3 Good Fit for Many Small Vertebrate Systems

The program is especially well suited to species with relatively simple age-structured demography and strong year-to-year demographic variation, such as many lizards, frogs, and small rodents. In such systems, a transparent age-based stochastic model can be more practical and easier to calibrate than a more complex individual-based framework.

8.4 Flexible Two-Species Extension

The app already supports a coexisting two-species mode with a shared effective carrying capacity. This is useful for comparative ecological questions and for extending the model toward coexistence, competition, or source-sink style scenarios.

8.5 Easy Customization

Because the program is written in Python with separate modules for model logic, summaries, plots, and validation, it can be modified relatively easily to add:

- more species
- alternative density-dependence rules
- different stochastic processes
- new export formats
- additional diagnostics

9. Comparison with VORTEX

VORTEX is a widely used population viability analysis tool. It is a strong program and in many cases will remain the better choice. The important point is that the two programs are designed for somewhat different purposes.

9.1 Advantages of This Program Relative to VORTEX

- simpler and more transparent Leslie-style structure
- easier to adapt to a specific research question
- easier to inspect mathematically

- faster to customize for particular small-vertebrate or age-structured scenarios
- direct emphasis on reproductive population summaries
- direct support for emigration-based density dependence
- straightforward browser interface and CSV export
- direct support for metapopulation occupancy comparisons
- easier modification of local and spatial demographic rules in code

Relative to VORTEX and other broad population viability analysis programs, the main advantage of this program is not that it is universally better, but that it is more transparent and easier to tailor when the research question is centered on Leslie-style age structure, stochastic variation, catastrophes, migration, patch occupancy, and comparison between species.

9.2 Advantages of VORTEX Relative to This Program

VORTEX is usually stronger when a study needs additional biological realism beyond the present Leslie-based structure, for example:

- explicit sex structure
- mating system complexity
- genetics and inbreeding
- harvest and supplementation
- more elaborate catastrophe modeling
- more elaborate metapopulation structure
- more elaborate dispersal among populations
- management scenarios used in formal population viability analysis

9.3 Recommended Interpretation

A fair interpretation is:

This program is especially useful when a researcher wants a focused, transparent, customizable stochastic Leslie-style model. VORTEX is often preferable when the question requires a broader individual-based population viability analysis framework with richer biological and management detail.

10. Current Limitations

Users should keep the following limitations in mind.

- maximum of 20 yearly age classes

- no explicit sex structure
- no genetics or inbreeding
- no direct density-dependent changes in fecundity or survival
- no built-in parameter fitting from observed time series
- no direct import yet from `Leslie.xlsx`
- no patch-specific catastrophe process yet
- no explicit distance-based dispersal kernel yet
- no explicit colonization cost, settlement cost, or patch-specific mortality during movement
- no individual-based mating or genetic processes

These limitations do not make the program weak. They simply define the scope of the current version.

11. Suggested Uses

This app is well suited for:

- exploring stochastic dynamics of age-structured populations
- comparing alternative fecundity and survival schedules
- testing sensitivity to carrying capacity and its variation
- studying persistence under repeated stochastic conditions
- comparing isolated and coexisting species scenarios
- generating intuitive teaching and research demonstrations

12. File Overview

Main files in the folder:

- `app.py`: web interface
- `model.py`: simulation engine
- `stochastic.py`: random annual draws
- `summaries.py`: replicate summaries
- `plots.py`: figure generation
- `validators.py`: input checks
- `defaults.py`: starter parameter values
- `io\_utils.py`: export helpers
- `launch\_app.bat`: launcher

13. Future Improvements

Possible future extensions include:

- import of parameter tables from Excel
- direct saving and reloading of scenario files
- alternative density-dependence functions
- explicit source and sink patches
- more than two species
- confidence envelopes based on quantiles
- sensitivity and elasticity analysis
- parameter-sweep automation
- report generation directly from the app

14. Citation and Copyright

© David Tarkhnishvili. Please cite this program if it is used in research, teaching, reports, or derivative software.

Suggested citation:

Tarkhnishvili, D. 2026. LESLIE: Leslie Population Dynamics App. Python/Streamlit application for stochastic age-structured population simulation.

15. Summary

The Leslie Population Dynamics App is a flexible and transparent stochastic age-structured simulation tool. Its main strengths are clarity, direct control of age-based demographic parameters, browser-based accessibility, and adaptability to customized ecological questions. It is particularly valuable when a simple but biologically meaningful Leslie-style structure is preferred over a more general and complex population viability framework.